

FIN Local Research — Release 10.40

Checkpoint P451–P453

Coherence Advantage, Global Coarse-Erasure Symmetry, and Canonical
Jordan Representation

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Central result. An exact-rational coherent three-slot causal tester has certified half-distance at least 0.52332810026048937, exceeding the proved global diagonal optimum by at least 0.00056473095851463685.

Additional theorems. The actual coarse three-use erasure objective is globally reduced to the palindromic line, and the P429 signed-moment optimizer is globally unique under the minimum-negative-mass rule.

Boundary. No full 21-dimensional optimum, selector, dimensional source, laboratory record, complete legacy-strict bridge, role transfer, or physical closure is claimed.

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Publication statement

This monograph reports local analytical and computer-assisted Programs P451, P452, and P453. It uses no laboratory record, external audit, internet result, remote computation, or physical validation.

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Chapter 1

Executive summary

This checkpoint completes local analytical and computational Programs P451, P452, and P453. It uses no laboratory data, external audit, external AI opinion, internet search, remote computation, or empirical record.

Three previously open mathematical boundaries are advanced.

1. **P451 — [Proven / Computer-assisted proof]**. The complete diagonal face of the three-slot causal tester cone is reduced by concavity and finite symmetry to one variable. Its global maximizer is algebraic,

$$a_* = 0.30884079156611205\dots,$$

and its half-distance is

$$D_{\text{diag}}^{\max} = 0.5227633693019748\dots$$

A separate exact-rational, strictly positive coherent tester has certified half-distance

$$0.52332810026048937 \leq D_{\text{coh}} \leq 0.52332810026049204.$$

It exceeds the global diagonal upper bound by at least

$$0.00056473095851463685.$$

Therefore diagonal optimality is rigorously refuted. Eight deterministic full-cone starts converge near 0.5233281002704, but no matching 21-dimensional dual certificate is yet available. The full causal optimum remains open.

2. **P452 — [Proven / Computer-assisted proof]**. The coarse-graining gap in P446/P448 is localized to the two-survivor sector. A new analytic inequality proves that reversal symmetrization cannot decrease this sector whenever

$$4q^2 \cos^2 \theta \geq 1.$$

At the declared parameters $q = 4/5$, $\theta = 2\pi/15$, exact rational trigonometric enclosures give a positive margin

$$4q^2 \cos^2 \theta - 1 = 1.1364871761393385\dots > 0.$$

Hence the complete four-sector probability simplex reduces to the P446 palindromic line. The old line certificate is now a full-simplex theorem:

$$0.46327828319203235 \leq D_{\text{coarse}}^{\max} \leq 0.4642707484333634,$$

with gap $0.0009924652413310642 < 10^{-3}$. This improves the P448 global upper bound by 0.0023597549471144896 .

3. **P453 — [Proven, conditional on inherited certified inputs]**. P429 gave a locally unique seven-contact primal–dual solution, while P430 proved its dual polynomial globally feasible. P453 adds the missing strict complementarity argument. Every global primal optimizer must place its positive Jordan measure on the five zero contacts of the dual and its negative Jordan measure on the two -1 contacts. The seven nodes are distinct, with minimum certified separation greater than 0.05815515198 , and their Vandermonde determinant is bounded away from zero by $1.4954513576571674 \times 10^{-9}$. The weights are therefore unique. The P429 measure is the unique global minimum-negative-mass representation and, at fixed signed mass, the unique minimum-total- variation representation.

The checkpoint constructs three typed objects:

- **O167 — Causal-Coherence Advantage Witness;**
- **O168 — Coarse-Erasure Symmetrization Certificate;**
- **O169 — Strict-Complementarity Jordan Gauge Fixer.**

These are mathematical and operational-design results. They do not provide a selector, canonical physical orientation, dimensional unit, apparatus, laboratory record, completed legacy-to-strict bridge, physical-role transfer, Standard Model, gravity theory, L_{total} , or Theory of Everything.

Chapter 2

Confidence convention

Label	Meaning in this checkpoint
[Proven]	Analytic consequence of stated premises, with exact symbolic or finite algebraic verification
[Computer-assisted proof]	Finite certificate using rational intervals, exact root isolation, or exhaustive interval branching
[Strong evidence]	Deterministic numerical evidence with falsification tests, but no complete proof certificate
[Conditional]	Correct only after an explicitly supplied mathematical or operational premise
[Open]	The present artifacts do not decide the question
[Blocked by external evidence]	A real laboratory, custody, calibration, or independent record is indispensable

Chapter 3

Inherited assumptions and guardrails

3.1 Accepted mathematical inputs

The following local results are used as working facts unless explicitly falsified here.

- The reduced one-qubit phase/dephasing Choi family is supplied at the stated dimensionless parameters.
- P445 proves the exact two-slot diagonal-support comb reduction.
- P446 proves a 10^{-3} interval upper/lower certificate on the palindromic three-use erasure-code line.
- P448 proves concavity of the fine-grained erasure majorant and a global full-simplex upper bound for that majorant.
- P449 proves the exact recursive compressed causal-support cone

$$\mathcal{C}_n = \{B_0 \oplus B_1 : B_0, B_1 \succeq 0, B_0 + B_1 \in \mathcal{C}_{n-1}\}.$$

- P429 certifies one unique seven-contact KKT solution in a declared exact rational box.
- P430 certifies the associated dual polynomial globally inside $[-1, 0]$ on $[0, 1]$.
- P450 proves that twelve moments alone admit exact null-cycle representation gauge freedom.

3.2 Kernel split

The historical/intermediate bridge kernel and strict working kernel remain distinct:

$$K_{\text{legacy,ont}}(d) = \frac{\alpha_{\text{geo}} \cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d},$$

$$K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}.$$

The locally researched Kernel new Legacy / Legacy* instance remains a historical intermediate subclass, not the strict kernel. None of P451–P453 substitutes one kernel for the other. The programs consume already typed downstream reduced-channel or strict-moment objects.

No legacy electroweak, electromagnetic, gravitational-hierarchy, or other physical role is transferred.

3.3 Selector, units, and physics

- QW-2191 remains open.

- Chronological slot order is a supplied causal interface, not a strict source of Z12 orientation.
- All channel parameters and times in this checkpoint are dimensionless.
- A mathematical tester is not an apparatus.
- A canonical signed measure is not a physical preparation.
- Synthetic computations are not laboratory evidence.
- The internal ontology remains

nadsoliton $\rightarrow$ light $\rightarrow$ matter $\rightarrow$ emergent observer,

with no lower informational layer beneath the nadsoliton.

Chapter 4

Current state map before P451–P453

Lane	Inherited state	Question attacked here	Outcome
Three-slot causal discrimination	P449 coherent echo refutes GHZ; global optimum open	Is the diagonal face globally solvable, and can coherence beat it?	Diagonal face solved; coherent advantage certified; full cone open
Three-use erasure code	P446 line proof; P448 looser full-simplex majorant	Can the actual coarse objective be symmetrized?	Yes under an explicit analytic condition; full-simplex 10^{-3} theorem
Signed-moment representation	P430 global value; P450 representation gauge	Is the minimum-negativity optimizer globally unique?	Yes by strict complementarity plus Vandermonde injectivity
Selector/orientation	QW-2191 open	Not imported	Unchanged
Dimensional source	No canonical unit	Not imported	Unchanged
Laboratory evidence	None admitted	Not fabricated	Unchanged
Legacy-to-strict bridge	Incomplete	Not replayed	Unchanged

Chapter 5

Program P451 — diagonal globality and causal coherence

5.1 Exact question

For three uses of the reduced memoryless phase/dephasing channel with

$$q = \frac{4}{5}, \quad \theta = \frac{\pi}{8},$$

determine the global optimum on the full diagonal history simplex. Then test whether an admissible off-diagonal normalizer in the exact P449 cone can strictly exceed that optimum.

5.2 Object type and objective

Let Δ_3 be the compressed three-use process difference on the eight diagonal Choi histories. For $N \in \mathcal{C}_3$, define

$$D(N) = \frac{1}{2} \left\| N^{1/2} \Delta_3 N^{1/2} \right\|_1.$$

The nuclear-norm variational formula makes D concave in N . This is the same mathematical half-distance used in P445 and P449.

5.3 Global reduction of the diagonal face

A diagonal N is a probability law on eight binary histories. The memoryless process is invariant under the six permutations of the three slots and global bit complement. Averaging a diagonal law over this finite group cannot lower the concave invariant objective.

The averaged law has the form

$$t_{000} = t_{111} = a, \quad t_h = b = \frac{1-2a}{6} \text{quad}(h \neq 000, 111),$$

where $0 \leq a \leq 1/2$. Thus every diagonal tester is dominated by one point on this line.

5.4 Exact one-variable formula

Block diagonalization under the same finite symmetry gives

$$D_{\text{diag}}(a) = \sqrt{R(a)} + \frac{12 - 24a}{125} \sqrt{2 - \sqrt{2}},$$

with

$$R(a) = \frac{16}{15625} \left[(-3520 + 1568\sqrt{2})a^2 + (1532 - 414\sqrt{2})a + 242 - 121\sqrt{2} \right].$$

The squared stationary equation is, up to the nonzero factor $-2048/244140625$,

$$\begin{aligned} 0 = & (-8836992 + 5639168\sqrt{2})a^2 \\ & + (3415528 - 1972208\sqrt{2})a \\ & - 323159 + 149850\sqrt{2}. \end{aligned}$$

Squaring produces two algebraic roots. Only one obeys the positive unsquared derivative branch. It is

$$\begin{aligned} a_* = & -\frac{5177925}{28304679968} \sqrt{3504977 - 2461052\sqrt{2}} \\ & - \frac{413025\sqrt{2}}{3538084996} \sqrt{3504977 - 2461052\sqrt{2}} \\ & + \frac{3691\sqrt{2}}{58384} + \frac{31987}{116768}. \end{aligned}$$

Exact symbolic substitution gives zero residual. Nested rational square-root enclosures give

$$a_* = 0.30884079156611205\dots$$

and

$$D_{\text{diag}}^{\max} = 0.5227633693019748\dots$$

Concavity proves this stationary point is globally maximizing on the whole diagonal face. The success probability on that face is

$$p_{\text{succ,diag}}^{\max} = \frac{1 + D_{\text{diag}}^{\max}}{2} = 0.7613816846509874\dots$$

5.5 Exact-rational coherent witness

Define $N = B_0 \oplus B_1$ by

$$B_0 = \begin{pmatrix} A & u & u & 0 \\ u & B & 0 & -u \\ u & 0 & B & -u \\ 0 & -u & -u & C \end{pmatrix}, \quad B_1 = \begin{pmatrix} C & -u & -u & 0 \\ -u & B & 0 & u \\ -u & 0 & B & u \\ 0 & u & u & A \end{pmatrix},$$

where

$$A = \frac{143583}{500000}, \quad B = \frac{68777}{1000000}, \quad C = \frac{941}{12500}, \quad u = \frac{9493}{1000000}.$$

Both blocks are congruent. One eigenvector has eigenvalue $B > 0$, and Sylvester reduction of the remaining three-dimensional block gives the exact positive minors

$$AB - 2u^2 = \frac{4892545471}{2500000000000} > 0,$$

$$ABC - 2u^2(A + C) = \frac{355371546810313}{25000000000000000} > 0.$$

Therefore $B_0, B_1 \succ 0$. Moreover,

$$B_0 + B_1 = \text{diag}(A + C, 2B, 2B, A + C),$$

and $A + C + 2B = 1/2$. This lies exactly in $\mathcal{C}_2$, so the P449 recursion admits $N \in \mathcal{C}_3$. The floating minimum eigenvalue 0.0577247734 is only a diagnostic; positivity follows from the rational minors.

5.6 Trace-distance certificate

The exact trigonometric entries of Δ_3 are enclosed using rational Machin/Taylor intervals. Their midpoints are rounded to 18-decimal rational numbers. For positive N ,

$$N\Delta_3 \sim N^{1/2}\Delta_3N^{1/2},$$

so their eigenvalues agree. The rational midpoint characteristic polynomial has eight real roots, each isolated in a disjoint exact rational interval. The residual entry radius is at most

$$4.175005697635287 \times 10^{-17}.$$

Using $\|N\|_2 \leq \text{Tr } N = 1$, the paid half-nuclear perturbation is at most

$$1.3360018232432918 \times 10^{-15}.$$

The final certified value is

$$0.52332810026048937 \leq D(N) \leq 0.52332810026049204.$$

Hence

$$D(N) - D_{\text{diag}}^{\max} \geq 0.00056473095851463685 > 0.$$

P451 verdict: diagonal optimality is **[Refuted]**. The displayed coherent witness and its advantage are **[Computer-assisted proof]**. The complete 21-dimensional optimum is **[Open]**.

5.7 Full-cone falsification scout

An interior effect parameterization was used:

$$C_0 = N_1^{1/2} E_2 N_1^{1/2}, \quad B_0 = N_2^{1/2} E_3 N_2^{1/2},$$

with logistic spectral effects $0 \prec E_j \prec I$. Eight deterministic starts converge near 0.5233281002704, close to but slightly above the rational witness. This is [\[Strong evidence\]](#) for the location of a smooth interior optimum. It is not promoted: no exact KKT box and no matching comb dual are exported.

Chapter 6

Program P452 — closing the coarse-erasure symmetrization gap

6.1 Exact question

P448 could symmetrize a fine-grained upper majorant, but the operational P446 objective discards the computational values of lost bits. Does the actual coarse objective still obey reversal symmetrization on the complete four-sector simplex?

6.2 Localizing the only gap

Let s be the number of surviving qubits.

- For $s = 3$, nothing is lost; coarse and fine forms coincide.
- For $s = 1$, the output is a two-dimensional skew block. All lost-value amplitudes have the same sign, so absolute value commutes with their sum; coarse and fine forms again coincide.
- Only $s = 2$ is nontrivial. If $v_0, v_1 \in \mathbb{R}^3$ are the skew-block vectors for the two lost values, then

$$D_{2,\text{coarse}} = \|v_0 + v_1\|_2, \quad D_{2,\text{fine}} = \|v_0\|_2 + \|v_1\|_2.$$

Their exact disclosure defect is

$$\delta_2 = \frac{2(\|v_0\|\|v_1\| - v_0 \cdot v_1)}{\|v_0\| + \|v_1\| + \|v_0 + v_1\|}.$$

This explains why the P448 majorant can be strict without introducing a new channel or a hidden physical effect: it reveals a classical label that the coarse experiment discards.

6.3 Analytic two-survivor inequality

Write the square roots of the four sector probabilities as

$$(x, y, z, w) = (\sqrt{p_0}, \sqrt{p_1}, \sqrt{p_2}, \sqrt{p_3}).$$

Set

$$x = \sqrt{2}X \cos \alpha, \quad w = \sqrt{2}X \sin \alpha,$$

$$y = \sqrt{2}Y \cos \beta, \quad z = \sqrt{2}Y \sin \beta,$$

with $\alpha, \beta \in [0, \pi/2]$ and $r = X/Y \geq 0$. The reversal average has square-root masses (X, Y, Y, X) . After the S_2 irreducible block reduction, the difference

$$D_2(\text{sym } p)^2 - D_2(p)^2$$

is a nonnegative scale times a quadratic polynomial in r . Its constant coefficient is

$$\frac{1}{9} \cos^2(2\beta) \geq 0.$$

Its linear coefficient is proportional to

$$1 - \sin(2\beta) \cos(\alpha - \beta) \geq 0.$$

Its quadratic coefficient is

$$Q = -\frac{1}{3} \cos(2\alpha) \cos(2\beta) + \gamma \cos^2(\alpha + \beta),$$

where

$$\gamma = \frac{4q^2 \cos^2 \theta}{3}.$$

If $\cos(2\alpha) \cos(2\beta) \leq 0$, then $Q \geq 0$ immediately. If the product is nonnegative, use

$$\cos^2(\alpha + \beta) - \cos(2\alpha) \cos(2\beta) = \sin^2(\alpha - \beta) \geq 0.$$

Thus $Q \geq 0$ whenever $\gamma \geq 1/3$, equivalently

$$4q^2 \cos^2 \theta \geq 1.$$

At $q = 4/5$, $\theta = 2\pi/15$, exact rational cosine intervals certify the strict margin $1.1364871761393385 \dots$

Therefore

$$D_2(p) \leq D_2\left(\frac{p + p^{\text{rev}}}{2}\right).$$

The $s = 1, 3$ components obey the same inequality by the inherited concavity argument. Hence the complete coarse objective satisfies

$$D_{\text{coarse}}(p) \leq D_{\text{coarse}}\left(\frac{p + p^{\text{rev}}}{2}\right).$$

6.4 Global full-simplex certificate

The reversal average is exactly

$$p(a) = (a, 1/2 - a, 1/2 - a, a).$$

P446 already exhaustively covered $0 \leq a \leq 1/2$ using outward-rounded binary64 interval arithmetic. P452 supplies the missing theorem that licenses the reduction of every full-simplex point to that line. Consequently

$$0.46327828319203235 \leq \max_{p \in \Delta_3} D_{\text{coarse}}(p) \leq 0.4642707484333634.$$

The gap is

$$0.0009924652413310642 < 10^{-3}.$$

This is a stricter operational upper bound than P448:

$$0.4666305033804779 - 0.4642707484333634 = 0.0023597549471144896.$$

P452 verdict: the full-simplex reversal theorem is **[Proven]** and the numerical global bracket is **[Computer-assisted proof]**. Exact optimizer uniqueness and unrestricted adaptive-code optimality remain **[Open]**.

Chapter 7

Program P453 — global uniqueness of the minimum-negativity measure

7.1 Exact question

P430 proved the value of the signed-moment optimization globally but did not claim every global optimizer was identical. P450 then showed that the moment vector alone admits exact null-cycle representation freedom. Does the minimum-negative-mass variational problem itself remove that freedom?

7.2 Primal and dual objects

Let

$$\mu = \mu_+ - \mu_-$$

be a finite signed measure on $[0, 1]$ reproducing the twelve strict moments. Its Jordan negative mass is

$$N(\mu) = \mu_-([0, 1]).$$

Let p be the P429/P430 dual polynomial, satisfying

$$-1 \leq p(x) \leq 0.$$

Weak duality is the identity and inequality

$$\begin{aligned} N(\mu) - \int p \, d\mu &= \int (-p) \, d\mu_+ + \int (1 + p) \, d\mu_- \\ &\geq 0. \end{aligned}$$

7.3 Equality forces the support

For any global primal optimizer, equality with the dual optimum holds. Both integrands on the right are nonnegative, so

$$\text{supp } \mu_+ \subseteq \{x : p(x) = 0\},$$

$$\text{supp } \mu_- \subseteq \{x : p(x) = -1\}.$$

P430 did more than show weak feasibility. Its curvature cells prove strict local extrema at the six interior contacts; its endpoint derivative proves the unique endpoint contact; and its Bernstein complement cells prove strict inequality everywhere else. Hence

$$\{p = -1\} = \{x_0, x_5\},$$

$$\{p = 0\} = \{x_1, x_2, x_3, x_4, 1\}.$$

Every global optimizer is therefore supported on these exact seven nodes, with the forced sign pattern

$$(-, +, +, +, +, -, +).$$

7.4 Vandermonde injectivity

The P429 node boxes are strictly ordered. Their minimum pairwise separation is certified above

$$0.058155151982592045.$$

For the seven-by-seven moment matrix

$$V_{ki} = x_i^k, \quad k, i = 0, \dots, 6,$$

the determinant is

$$\det V = \prod_{0 \leq i < j \leq 6} (x_j - x_i).$$

Interval multiplication gives

$$|\det V| > 1.4954513576571674 \times 10^{-9} > 0.$$

Thus the first seven moment equations already determine all seven weights uniquely. The remaining five equations are satisfied by the inherited P429 Krawczyk solution. No second global optimizer can exist.

7.5 Minimum total variation

The signed zeroth moment

$$m_0 = \mu([0, 1])$$

is fixed. Therefore

$$\|\mu\|_{\text{TV}} = m_0 + 2N(\mu).$$

Minimizing negative mass is equivalent to minimizing total variation. The unique optimum has

$$N_* = 0.7073534677231137\dots,$$

$$\|\mu_*\|_{\text{TV}} = 2.401532838636556\dots$$

7.6 What is and is not canonical

Within the declared variational problem, the result is canonical: there is one minimum-TV signed measure, hence one reduced atomic input to O163.

Moments alone are still insufficient. Removing the minimum-negativity rule restores the P450 gauge

$$\mu_\epsilon = \mu_* + \epsilon\nu_{12},$$

where ν_{12} annihilates moments zero through eleven. Thus P453 does not falsify P450; it supplies exactly the additional mathematical rule P450 proved necessary.

The rule is not an internally generated apparatus. A real implementation would still have to demonstrate that its preparation instrument realizes the seven atoms and signs.

P453 verdict: global optimizer uniqueness is [**Proven conditional on the inherited P429/P430 computer-assisted certificates and signed-moment duality**]. Physical preparation remains **[Blocked by external evidence]**.



Figure 7.1: The certified causal-coherence advantage, the global coarse-erasure palindromic reduction, and the unique minimum-TV signed measure.

Chapter 8

Newly constructed theoretical objects

8.1 O167 — Causal-Coherence Advantage Witness

Required field	Definition and audit
Name	Causal-Coherence Advantage Witness
Domain/codomain	Exact rational $N \in \mathcal{C}_3 \mapsto D(N) \in \mathbb{R}_{\geq 0}$
Complete definition	The displayed block matrix $B_0 \oplus B_1$ with rational A, B, C, u
Premises	P449 cone recursion; declared reduced $q = 4/5, \theta = \pi/8$ channel
Transformation law	Global complement exchanges the blocks and signs but preserves D
Generated theorem	A coherent causal tester exceeds the global diagonal optimum
Necessity test	Setting $u = 0$ returns to the diagonal face and removes the certified advantage
Removal attempt	The complete diagonal optimization proves no diagonal substitute can match the witness
Strict/legacy relation	Downstream reduced-channel object; derives neither kernel and substitutes neither
Selector dependence	None; chronological interface is supplied and does not discharge QW-2191
Dimensional status	Dimensionless
Operational interpretation	Mathematical three-slot tester normalizer
Failure mode	Does not establish the full 21-dimensional optimum or laboratory implementability
Confidence	[Computer-assisted proof] for feasibility and advantage

8.2 O168 — Coarse-Erasure Symmetrization Certificate

Required field	Definition and audit
Name	Coarse-Erasure Symmetrization Certificate

Required field	Definition and audit
Domain/codomain	$p \in \Delta_3 \mapsto (p + p^{\text{rev}})/2$ with a monotonicity theorem for D_{coarse}
Complete definition	The survivor-sector decomposition plus the nonnegative quadratic coefficients under $4q^2 \cos^2 \theta \geq 1$
Premises	Three-use Hamming-sector code; heralded loss; $q = 4/5, \theta = 2\pi/15$
Transformation law	Invariant under sector reversal
Generated theorem	Complete simplex reduces to the palindromic line
Necessity test	Without the $s = 2$ inequality, P448 only controls a strictly larger disclosed-label majorant
Removal attempt	Direct triangle inequality reproduces only the old loose P448 upper bound
Strict/legacy relation	Kernel-split robust downstream code theorem
Selector dependence	None; reversal averaging does not select an orientation
Dimensional status	Dimensionless
Operational interpretation	Exact mathematical control of lost-label coarse graining
Failure mode	The condition may fail for other q, θ ; unrestricted input families remain open
Confidence	[Proven] reduction; [Computer-assisted proof] value bracket

8.3 O169 — Strict-Complementarity Jordan Gauge Fixer

Required field	Definition and audit
Name	Strict-Complementarity Jordan Gauge Fixer
Domain/codomain	Strict moment vector plus minimum-negativity variational rule $\mapsto$ unique signed measure
Complete definition	Equality support under the P430 dual polynomial followed by seven-node Vandermonde inversion
Premises	P429 Krawczyk box; P430 global strict contact cells; inherited signed-measure duality
Transformation law	Coordinate changes must push forward moments, measure, and dual polynomial together
Generated theorem	Unique global minimum-negative-mass and minimum-TV representation
Necessity test	The P450 null cycle survives if the variational rule is removed
Removal attempt	Moments alone give an unbounded O163 risk gauge orbit
Strict/legacy relation	Uses strict moments; no legacy substitution or role transfer
Selector dependence	The coordinate chart is supplied; no canonical physical orientation is generated

Required field	Definition and audit
Dimensional status	Dimensionless
Operational interpretation	Canonical mathematical input for a sampler, not a preparation device
Failure mode	A different declared objective may select a different representation; physical realization remains external
Confidence	[Proven conditional on inherited certified inputs]

Chapter 9

Falsification attempts and failed approaches

9.1 P451: diagonal conjecture destroyed

The aesthetically simplest conjecture was that the group-symmetrized diagonal optimum might already solve the complete causal cone. It fails. The rational coherent witness lies strictly above its certified upper bound. This counterexample is not a numerical accident: positivity, causality, and the distance gap all have finite certificates.

9.2 P451: floating optimum not promoted

Eight full-cone starts agree to approximately ten decimal places. This could tempt a global claim. It is rejected because the objective is concave but is being maximized, the cone has 21 affine dimensions, and no supporting dual matrix or interval KKT inclusion has been constructed.

9.3 P452: fine majorant no longer used as equality

P448's revealed-label majorant can be strictly larger than the coarse objective. P452 does not call that majorant the channel value. It proves a separate inequality for the exact coarse two-survivor block and then reuses only the already valid P446 line cover.

9.4 P452: parameter universality rejected

The analytic condition $4q^2 \cos^2 \theta \geq 1$ is part of the theorem. No claim is made that reversal symmetrization holds for every coherence and phase.

9.5 P453: moments-alone canonicity remains refuted

P453 does not retract the P450 no-go. The canonical object arises only after adding the minimum-negative-mass variational rule. Without it, the exact finite-difference null cycle remains.

9.6 P453: mathematical representation is not preparation

Global uniqueness of a measure does not force a physical source to emit its atoms. Any such assertion would require apparatus, calibration, event-level records, custody, and an independently frozen protocol.

Chapter 10

Reproducibility and numerical boundaries

Pro-gram	Exact/finite method	Floating method and Decisive boundary role
P451	Nested radical intervals; exact ratio- nal PSD minors; rational characteris- tic polynomial root isolation; nuclear perturbation bound	Eight seeded L-BFGS-B No full-cone dual cer- starts only scout the full tificate cone
P452	Analytic nonnegative-coefficient proof; 256 seeded random points exact rational cosine margin; inherited outward 1D interval cover	are falsification only Exact maximizer/ uniqueness not proved
P453	Strict complementary support theo- rem; exact node boxes; rational Van- dermonde lower bound	Midpoints appear only in Inherits P429/P430 figures and duality assump- tions

The P451 characteristic polynomial has eight disjoint real root intervals. The midpoint-to-exact perturbation is paid after, rather than ignored. The P452 random audit found no positive symmetrization defect; the theorem does not depend on this audit. The P453 determinant uses lower node separations, not rounded midpoint nodes.

Chapter 11

Legacy/strict, selector, dimensional, and operational audit

Boundary	Result after P451–P453
K_legacy_ont versus K_strict_gate	Still separated; no substitution
Legacy* / Kernel new Legacy	Historical bridge subclass only; not an input to these proofs
Self-coupling diagram	Still intuition/provenance, not a typed feed-back state map
Legacy-to-strict completion	Open
Legacy physical-role transfer	Not started
QW-2191 selector	Open
Canonical orientation/phase origin	Not generated
Dimensional time/length/action	Not generated
State/preparation/instrument/apparatus	Mathematical interfaces only
Laboratory record	None
Independent custody/hold-out	None executed
Information-to-thermodynamic entropy	Not established
Standard Model / gravity / L_{total} / ToE	Not established

The dual functional calculus

$$U_t = e^{-itA}, \quad P_t = e^{-tA}$$

remains mathematical wave/heat duality. P451's causal coherence does not turn this duality into an observer, clock, environment, or record.

Chapter 12

Ranked next research programs

The following ranking is for decisive local analytical/computational progress, not physical plausibility. Probabilities are judgmental planning weights, not theorem probabilities.

Rank	Program	Exact target	Cheapest decisive method	Probability of decisive local result
1	P454	Matching primal/dual certificate for the full 21D P451 causal cone	Derive the nested comb SDP dual; solve the symmetry-reduced KKT system; rationalize and verify PSD slacks	0.72
2	P455	Exact optimum on the coherent symmetry face containing O167	Symbolic block diagonalization, algebraic stationarity, interval root isolation	0.81
3	P456	Decide whether O167 coherence advantage persists for $n = 4$	Extend P449 recursion; construct one rational witness and compare with the global diagonal face	0.69
4	P457	Tighten P452 full-simplex value from 10^{-3} to 10^{-6}	Re-run the licensed 1D outward branch cover with a smaller tolerance	0.94
5	P458	Prove or refute uniqueness of the P452 palindromic maximizer	Derivative interval isolation and monotonicity cells on $[0, 1/2]$	0.84
6	P459	Propagate P453 global uniqueness through the complete O163 detector box	Combine the unique contact support with P447 atom intervals and allocation sensitivity	0.91
7	P460	Compare minimum-TV gauge fixing with a representation-quotient estimator	Construct null-cycle-invariant estimators or prove a finite-dimensional no-go	0.63
8	P461	Formalize the P452 trigonometric/algebraic reduction	Lean proof of the coefficient signs, leaving only a typed cosine provider if necessary	0.58
9	P462	Formalize P453 complementarity and Vandermonde uniqueness	Lean finite-measure support lemma plus finite Vandermonde theorem	0.61
10	P463	Freeze a preparation-level executable contract for O169	Typed manifest, admission validator, synthetic integration test explicitly marked nonphysical	0.88

The selected next batch is

$$P454 \longrightarrow P455 \longrightarrow P457.$$

P454 attacks the most important open theorem. P455 supplies an exact fallback if the full dual cannot be closed. P457 is cheap, licensed by P452, and guarantees a durable quantitative improvement before a later risky batch.

Chapter 13

Exact artifact list

- FIN_Local_Research_Checkpoint_P451_P453_EN.md
 - FIN_Local_Research_Checkpoint_P451_P453_EN.tex
 - FIN_Local_Research_Checkpoint_P451_P453_EN.pdf
 - FIN_Current_Local_Research_Report_EN.pdf
 - FIN_Local_Research_Checkpoint_P451_P453_State.json
 - RELEASE_10_40_PROGRAMS_451_453.md
 - FIN_Checkpoint_P451_P453_AGENTS_Guardrail.txt
 - FIN_PROGRAMS_451_453_RELEASE_MANIFEST.sha256
 - FIN_Programs_451_453_Results.json
 - FIN_Programs_451_453_Summary.csv
 - FIN_Programs_451_453_P451_Causal_Coherence.csv
 - FIN_Programs_451_453_P451_Coherent_Witness.npz
 - FIN_Programs_451_453_P452_Coarse_Symmetrization.csv
 - FIN_Programs_451_453_P453_Canonical_Representation.csv
 - FIN_Programs_451_453_Figures/p451_p453_coherence_symmetry_and_gauge_fixing.png
 - fin_programs_451_453.py
 - fin_programs_451_453_to_latex.py
 - test_fin_programs_451_453.py
 - test_fin_checkpoint_p451_p453.py
-

Chapter 14

Restart instructions

Run the scientific batch:

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 fin_programs_451_453.py
```

Run current and inherited regression tests:

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 python3 -m unittest -v \  
  test_fin_programs_451_453.py \  
  test_fin_checkpoint_p451_p453.py \  
  test_fin_checkpoint_p448_p450.py \  
  test_fin_checkpoint_p445_p447.py
```

Regenerate and compile the monograph:

```
python3 fin_programs_451_453_to_latex.py  
lualatex -interaction=nonstopmode -halt-on-error \  
  FIN_Local_Research_Checkpoint_P451_P453_EN.tex  
lualatex -interaction=nonstopmode -halt-on-error \  
  FIN_Local_Research_Checkpoint_P451_P453_EN.tex
```

Verify the immutable package:

```
sha256sum -c FIN_PROGRAMS_451_453_RELEASE_MANIFEST.sha256
```

Before P454, reread the latest AGENTS.md guardrail and preserve all kernel, selector, dimensional, and physical-evidence boundaries.

Chapter 15

Final conclusion

P451–P453 resolve three precise mathematical ambiguities without promoting FIN to physics.

The three-slot reduced channel needs genuine causal coherence: the complete diagonal theory is solvable and is strictly suboptimal. The three-use heralded erasure code no longer has a full-simplex symmetry gap: its coarse objective is globally controlled by the palindromic line under a proved parameter condition. The strict signed-moment problem no longer has multiple minimum-negativity representations: strict complementarity forces seven contacts, and Vandermonde injectivity forces their weights.

The deepest surviving interpretation of this batch is therefore a family of finite, dimensionless operator-optimization structures with increasingly sharp causal, coarse-graining, and variational interfaces. The missing physics is still missing: no theorem here creates a physical clock, unit, preparation device, detector, observer, laboratory record, or unique strict orientation.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/fin-mpl-1038 \  
python3 fin_programs_451_453.py
```

```
python3 -m unittest -v \  
    test_fin_programs_451_453.py \  
    test_fin_checkpoint_p451_p453.py
```

```
lualatex -interaction=nonstopmode -halt-on-error \  
    FIN_Local_Research_Checkpoint_P451_P453_EN.tex
```

```
sha256sum -c FIN_PROGRAMS_451_453_RELEASE_MANIFEST.sha256
```

Suggested citation:

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